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***B.Tech. Degree II Semester Regular and I Semester Supplementary
Examination May 2016***

**IT/CS//EC/CE/EE/ME/SE 1104 A / 1204 B BASIC CIVIL ENGINEERING
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

**PART A
(Answer *ALL* Questions)**

(10 × 2 = 20)

- I. (a) Explain the various grades of ordinary Portland cement available in India.
(b) Explain the classification of bricks used for construction purpose.
(c) What are the functions of water in concrete?
(d) What are the conditions under which a raft foundation is preferred?
(e) Explain the different types of machine foundations that are commonly used.
(f) Differentiate English bond and Flemish bond.
(g) Explain the principle of “working from whole to part” in surveying.
(h) Briefly explain the method to range a line when its end points are inaccessible.
(i) Define Coverage and Floor Area Ratio.
(j) Explain briefly the concepts of intelligent buildings.

PART B

(4 × 10 = 40)

- II. Write short notes on : (10)
(i) Acid resistant cement
(ii) Blast furnace cement
(iii) Sulphate resisting cement
(iv) Rapid hardening cement
- OR**
- III. Explain various market forms of steel sections with neat sketches. (10)
- IV. Explain the classification of deep foundations based on the materials of construction. (10)
- OR**
- V. (a) Discuss briefly the various types of roof covering materials. (7)
(b) Write short notes on random rubble masonry. (3)

- VI. (a) What are the uses of reconnaissance survey? (4)
- (b) The following readings were taken with a dumpy level. The instrument was shifted after 4th and 7th reading. (6)
- 0.155, 2.050, 1.850, 1.500, 2.500, 2.105, 1.800, 0.990.
- Calculate the reduced levels of all the stations. Given the RL of starting station is +100.00.
- OR**
- VII. (a) Explain the temporary adjustments of a dumpy level. (6)
- (b) Explain the following terms. (4)
- (i) Back sight
 - (ii) Fore sight
 - (iii) Bench mark
 - (iv) Reduced level.
- VIII. (a) Explain the general provisions regarding site and building requirements as per the Kerala Municipal Building Rule 1999. (7)
- (b) Explain the qualities required for drinking water. (3)
- OR**
- IX. Explain the classification of urban and rural roads. (10)

